AS · Unit 2 · WPH12 Waves & Electricity

Progressive Waves

Drop a stone into a pond and the ripples spread out, but the water itself only bobs up and down. That is the whole idea of a wave: **energy travels, the medium stays where it is**. On this page we describe waves with a handful of quantities, link them with $v = f\lambda$, learn to read wave graphs, measure the speed of sound, see why waves get weaker as they spread, and use echoes to find things we can't see.

Describing a wave

A **progressive** (travelling) wave transfers energy from one place to another **without transferring matter**. Each particle of the medium just **oscillates** about its own rest (equilibrium) position and passes the disturbance on to its neighbour a moment later. The pattern moves; the particles don't. A cork on a pond goes up and down as the ripples pass, but it doesn't ride away on them.

A handful of quantities describe any wave. Learn the definitions closely, because one-mark "state what is meant by…" questions are common.

DEFINITION

The language of waves

- **Displacement**: the distance of a particle from its rest position, in a stated direction (so it can be negative).
- **Amplitude** A: the maximum displacement from the rest position.
- **Wavelength** λ: the distance between two adjacent points on the wave that are oscillating in phase, for example crest to crest. Unit: m.
- **Period** T: the time taken for one complete oscillation of a particle. Unit: s.
- **Frequency** f: the number of complete oscillations per second. Unit: hertz, Hz (1 Hz = 1 s^{-1}).
- **Wave speed** v: the distance travelled by the wave (by the energy) per unit time. Unit: m s^{-1}.

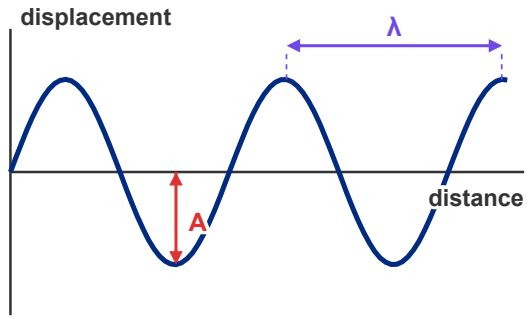
Period and frequency

$$f = \frac{1}{T}$$

f: frequency / Hz T: period / s

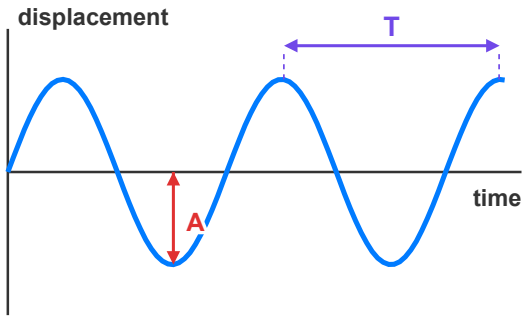
If a particle makes 50 oscillations every second, each one takes $\frac{1}{50}\ \text{s} = 0.020\ \text{s}$. Frequency and period are two ways of saying the same thing.

Two points are **in phase** when they are doing exactly the same thing at the same moment: same displacement, moving the same way. Points one whole wavelength apart are always in phase, which is why the definition of λ uses that idea. Phase difference is covered fully on the **Superposition & Stationary Waves** page.



a snapshot of every particle at one instant

Displacement–distance graph: the repeat length along the x-axis is the **wavelength**.



the motion of one particle over time

Displacement–time graph: the repeat length along the x-axis is the **period**.

EXAM TIP

The two graphs above look identical. Before you read anything off a wave graph, **check the label on the x-axis**. Distance means you can read λ; time means you can read T (and then $f = 1/T$). Mixing the two up is the most common way to lose marks here.

Transverse and longitudinal waves

Waves are sorted by how the particles move compared with the direction the wave travels, which is the direction of **energy transfer**.

DEFINITION

Transverse and longitudinal waves

In a **transverse wave** the particles oscillate **perpendicular** to the direction of energy transfer.

In a **longitudinal wave** the particles oscillate **parallel** to the direction of energy transfer. This produces **compressions** (particles closer together than normal, so the pressure is higher than normal) and **rarefactions** (particles further apart than normal, so the pressure is lower than normal).

Transverse waves

Flick one end of a long rope up and down and a hump runs along it. Each bit of rope only moves up and down, at right angles to the way the hump travels. Waves on a guitar string, S-waves in an earthquake, and **all electromagnetic waves** (radio, light, X-rays…) are transverse. EM waves are oscillating electric and magnetic fields, so they need no medium at all and cross the vacuum of space at $c = 3.00 \times 10^{8}\ \text{m s}^{-1}$. Only transverse waves can be **polarised**, which is how we know light is transverse (see Refraction, Diffraction & Polarisation).

Longitudinal waves: pressure and molecules

A loudspeaker cone pushes forward and squashes the air molecules in front of it together: a **compression**. Then it pulls back, leaving a region where the molecules are spread out: a **rarefaction**. Each molecule only shuffles backwards and forwards about its own average position, passing the push on to the next one. Sound, ultrasound and earthquake P-waves are longitudinal. Sound needs a medium: in a vacuum there are no molecules to push, which is why space is silent.

Here is the subtle part. Where is the **pressure** highest? At the centre of a compression. The molecules just to its left have been pushed right and the ones just to its right have been pushed left, so they crowd together. But the molecule *at* the centre has been pushed neither way: its **displacement is zero**. So the pressure variation is a quarter of a wavelength out of step with the displacement. The particles with the biggest displacement sit where the pressure is normal.

	Transverse	Longitudinal
Particle oscillation	perpendicular to energy transfer	parallel to energy transfer
Features	crests (peaks) and troughs	compressions and rarefactions
Examples	EM waves, waves on strings, S-waves	sound, ultrasound, P-waves
Can be polarised?	yes	no
Distance from a crest (or compression) to the next trough (or rarefaction)	$\lambda/2$	$\lambda/2$

COMMON MISTAKE

A displacement–distance graph of a sound wave is **not a picture** of the wave. It looks like a transverse wave only because we plot the sideways displacement upwards. The crests of the graph are **not** compressions: they are the molecules that have moved furthest, and the pressure there is normal.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Why light travels at exactly 3.00 × 10⁸ m s⁻¹

The speed of a mechanical wave depends on the medium: how strongly it pulls displaced particles back, and how much inertia they have. A wave on a string is a good example, with $v = \sqrt{T/\mu}$ (on the Superposition & Stationary Waves page). In 1865 James Clerk Maxwell showed that a changing electric field makes a magnetic field and vice versa, so the two can keep each other going through empty space. His equations predicted the speed of such a wave from two constants of electricity and magnetism:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.85 \times 10^{-12})}} = 3.00 \times 10^{8}\ \text{m s}^{-1}$$

That matched the measured speed of light, and it was how we discovered that light is an electromagnetic wave. Here μ_0 is the permeability of free space, which you won't meet in the IAL course.

The wave equation v = fλ

The wave equation

FORMULA SHEET

$$v = f\lambda$$

v: wave speed / m s⁻¹ f: frequency / Hz λ: wavelength / m

This works for every wave, transverse or longitudinal, sound or light. It comes straight from speed = distance ÷ time.

Deriving v = fλ

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Speed is distance travelled divided by the time taken:

$$v = \frac{\text{distance}}{\text{time}}$$

2. Watch one crest. In one period T the source makes exactly one complete oscillation, so it sends out exactly one more wavelength. The crest moves forward by exactly one wavelength λ in that time:

$$v = \frac{\lambda}{T}$$

After a time T, the particle one wavelength ahead is doing exactly what this one was doing, so the whole pattern has shifted along by λ.

3. Since $f = 1/T$, dividing by T is the same as multiplying by f:

$$\boxed{v = f\lambda}$$

KEY IDEA

Who decides what?

- The **source** sets the frequency. A tuning fork makes 440 oscillations per second whatever it is sitting in.
- The **medium** sets the speed. Sound in air at room temperature travels at about 340 m s^{-1} whatever note you play.
- The **wavelength** is whatever fits: $\lambda = v/f$. Raise the frequency in the same medium and the waves bunch up; the speed doesn't change.

When a wave crosses into a new medium, its **frequency stays the same** (every crest that arrives at the boundary must leave it), but its speed changes, so its wavelength changes too.

WORKED EXAMPLE

A loudspeaker emits a note of frequency 440 Hz. The speed of sound is 340 m s^{-1} in air and 1500 m s^{-1} in water.
(a) Calculate the wavelength of the sound in air.
(b) The sound passes into a swimming pool. State its frequency in the water and calculate its wavelength there.

Show solution

(a) Rearrange $v = f\lambda$:

$$\lambda = \frac{v}{f} = \frac{340}{440} = 0.773 \text{ m } (3 \text{ s.f.})$$

(b) The frequency is set by the source and doesn't change at the boundary: $f = 440$ Hz. The speed increases, so the wavelength increases in proportion:

$$\lambda = \frac{1500}{440} = 3.41 \text{ m } (3 \text{ s.f.})$$

Displacement–distance and displacement–time graphs

There are two ways to graph a wave, and the spec expects you to **draw and interpret** both, for transverse and longitudinal waves.

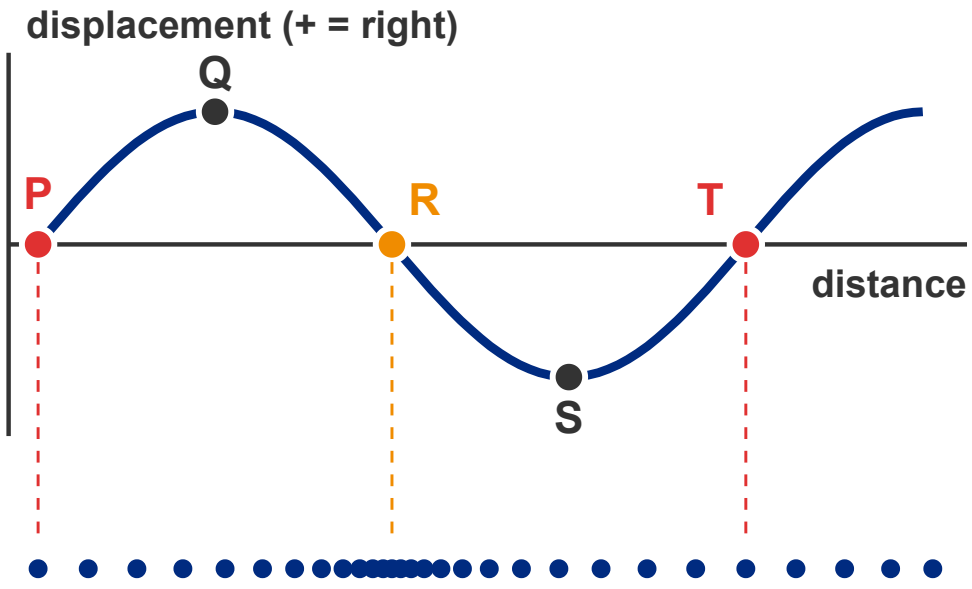
- A **displacement–distance** graph is a snapshot: the displacement of *every* particle along the wave at *one* instant. Read the **amplitude** and the **wavelength** from it.
- A **displacement–time** graph follows *one* particle over time. Read the **amplitude** and the **period** from it, then $f = 1/T$.
- Put the two together and you have everything: $v = f\lambda = \lambda/T$.

Which way is a particle moving?

On a displacement–distance graph, imagine sliding the whole curve a tiny distance in the direction the wave travels. Whatever the curve does at a particle's position tells you which way it moves next. For a wave travelling to the right, a particle on the slope just *in front* of a crest is moving **up** (the crest is about to arrive), and one just *behind* a crest is moving **down**. Particles at a crest or trough are momentarily at rest.

Finding compressions on a longitudinal graph

For a longitudinal wave, take positive displacement to mean "displaced to the right". Where the graph crosses zero **sloping downwards**, the particles on the left have moved right and those on the right have moved left: they bunch together, so that is the centre of a **compression**. Where it crosses zero **sloping upwards**, they have moved apart: a **rarefaction**.



The same instant shown two ways. R is the centre of a compression (highest pressure) and P and T are centres of rarefactions (lowest pressure). At Q and S the particles have their largest displacements, and the pressure there is normal. P to T is one wavelength.

WORKED EXAMPLE

A sound wave has frequency 500 Hz. On its displacement–distance graph, the centre of a compression and the centre of the nearest rarefaction are 0.34 m apart. Calculate the speed of the sound.

Show solution

A compression and the next rarefaction are half a wavelength apart (R to T in the figure), so

$$\lambda = 2 \times 0.34 = 0.68 \text{ m}$$

$$v = f\lambda = 500 \times 0.68 = 340 \text{ m s}^{-1}$$

WORKED EXAMPLE

A displacement–distance graph for a wave on a rope shows crests 1.2 m apart and a maximum displacement of 3.0 cm. A displacement–time graph for one point on the rope shows the pattern repeating every 0.080 s.

(a) Calculate the frequency and the speed of the wave.
(b) How far does a crest travel in 0.20 s?

Show solution

(a) The period comes from the displacement–time graph, the wavelength from the displacement–distance graph:

$$f = \frac{1}{T} = \frac{1}{0.080} = 12.5 \text{ Hz}$$

$$v = f\lambda = 12.5 \times 1.2 = 15 \text{ m s}^{-1}$$

(b) The crest moves at the wave speed: $15 \times 0.20 = 3.0$ m. The amplitude (3.0 cm) isn't needed: it doesn't affect the speed.

Core Practical 4: the speed of sound

CORE PRACTICAL

Determine the speed of sound in air using a 2-beam oscilloscope, signal generator, speaker and microphone

Connect a **signal generator** to a **loudspeaker** and also to channel 1 of a **dual-beam oscilloscope**. Connect a **microphone** to channel 2. Lay a metre rule along the line from the speaker to the microphone. Set a frequency of a few kHz and adjust the time-base until you can see a few cycles of each trace.

1. Start with the microphone close to the speaker. Move it slowly along the rule until the two traces are **in phase** (peaks line up). Record its position.
2. Move the microphone further away. The traces drift out of phase and come back **in phase every time the microphone moves one wavelength**, because the sound then takes exactly one more period to arrive.
3. Record the position each time they are back in phase (or count 10 wavelengths and measure the total distance).
4. Plot **microphone position against number of wavelengths moved,** n. It's a straight line with **gradient =** λ. Then $v = f\lambda$.
5. Repeat for several frequencies. Plotting λ against $1/f$ gives a straight line through the origin with **gradient =** v.

- **Measure many wavelengths and divide**: judging "exactly in phase" is uncertain by a few millimetres, so spreading that over 10 wavelengths shrinks the percentage uncertainty.
- Line up the points where the traces cross the centre line rather than the peaks: crossings are sharper and easier to judge.

- **Reflections** from walls and the bench add extra sound that distorts the microphone trace. Work in a large space, away from hard surfaces.
- The frequency on the signal generator dial may be inaccurate. Check it with the oscilloscope: T = number of divisions for one cycle × time-base setting, and $f = 1/T$.
- Keep the microphone on the line from the speaker and measure from the same point on its stand each time.
- Record the room temperature, since the speed of sound depends on it (about 0.6 m s^{-1} faster for every 1 °C warmer) NOT IN SPEC.
- **Safety:** keep the volume low, as prolonged loud high-pitched sound is uncomfortable and can damage hearing.

WORKED EXAMPLE

In Core Practical 4 the signal generator is set to 2.50 kHz. The microphone positions at which the traces are in phase are 12.0, 25.8, 39.5, 53.4 and 67.0 cm. Find the speed of sound.

Show solution

Successive positions are one wavelength apart. Plotting position against $n = 0, 1, 2, 3, 4$ gives a best-fit gradient of 13.76 cm. Without a graph, use the first and last readings (four wavelengths apart), which uses all the spacing at once:

$$\lambda = \frac{67.0 - 12.0}{4} = 13.75 \text{ cm} = 0.1375 \text{ m}$$

$$v = f\lambda = 2500 \times 0.1375 = 344 \text{ m s}^{-1}$$

This is close to the accepted 343 m s^{-1} at 20 °C. Averaging the four separate gaps (13.8, 13.7, 13.9, 13.6 cm) gives the same answer, because the middle readings cancel out.

Intensity

Stand close to a speaker and it's loud; walk away and it gets quieter, even though the speaker hasn't changed. The power hasn't gone anywhere. It has been **spread over a bigger area**. That idea is intensity.

DEFINITION

Intensity

The **intensity** of radiation is the **power per unit area**, where the area is perpendicular to the direction the wave is travelling. Unit: W m^{-2}.

Intensity of radiation

FORMULA SHEET

$$I = \frac{P}{A}$$

I: intensity / W m⁻² P: power / W A: area / m²

For a **point source** that sends its power out equally in all directions, the energy at distance r is spread over the surface of a sphere of radius r. That gives the inverse square law, which is just $I = P/A$ with the right area.

Intensity at a distance r from a point source

1. A point source radiates power P equally in every direction. At a distance r all of that power passes through an imaginary sphere of radius r, whose surface area is

$$A = 4\pi r^2$$

2. Apply the definition of intensity to that sphere:

$$\boxed{I = \frac{P}{4\pi r^2}}$$

3. So $I \propto 1/r^2$. Double the distance and the same power covers $2^2 = 4$ times the area, so the intensity falls to a quarter. Triple it and the intensity falls to a ninth.

This assumes nothing is absorbed on the way and the source really does radiate evenly in all directions.

A parallel beam, like a laser, hardly spreads, so its area and therefore its intensity barely change with distance. That is why a 1 mW laser pointer can be dangerous to your eye while a 1 mW glow from a bulb is not: the same power goes into a tiny area. Intensity also grows with the square of the amplitude, so doubling the amplitude of a wave carries four times the power NOT IN SPEC.

WORKED EXAMPLE

A small loudspeaker emits 0.50 W of sound evenly in all directions.

(a) Calculate the intensity 3.0 m from the speaker.

(b) An eardrum has an area of $6.0 \times 10^{-5}\ \text{m}^2$. Calculate the sound power entering one ear at this distance.

(c) At what distance would the intensity fall to $1.0 \times 10^{-4}\ \text{W m}^{-2}$?

Show solution

(a) The power spreads over a sphere of radius 3.0 m:

$$I = \frac{P}{4\pi r^2} = \frac{0.50}{4\pi \times 3.0^2} = 4.42 \times 10^{-3}\ \text{W m}^{-2}$$

(b) Rearrange $I = P/A$ for the small area of the eardrum:

$$P = IA = 4.42 \times 10^{-3} \times 6.0 \times 10^{-5} = 2.7 \times 10^{-7}\ \text{W}$$

(c) Rearrange for r:

$$r = \sqrt{\frac{P}{4\pi I}} = \sqrt{\frac{0.50}{4\pi \times 1.0 \times 10^{-4}}} = \sqrt{398} = 20\ \text{m}$$

Check with ratios: the intensity has dropped by a factor of 44, so the distance has grown by $\sqrt{44} = 6.6$, and $3.0 \times 6.6 = 20$ m.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Decibels: how your ears measure intensity

Your ears respond to an enormous range: the loudest sound you can bear is about 10^{12} times the intensity of the quietest you can hear. So sound levels are quoted on a logarithmic scale, in decibels:

$$L = 10 \log_{10}\left(\frac{I}{I_0}\right)\ \text{dB}, \qquad I_0 = 1.0 \times 10^{-12}\ \text{W m}^{-2}$$

Every +10 dB means 10 times the intensity, and +3 dB roughly doubles it. Moving twice as far from a point source quarters the intensity, which is a drop of about 6 dB. The speaker in the example above gives $10 \log_{10}(4.42 \times 10^{-3}/10^{-12}) = 96$ dB at 3.0 m.

Reflection, transmission and pulse-echo

When a wave reaches an **interface** (a boundary between two media), some of it is **reflected** and the rest is **transmitted** into the second medium. You hear an echo off a wall, but some of the sound also goes into the wall. You see a faint reflection in a shop window and the shop behind it.

- The **reflected** wave stays in the same medium, so its speed, frequency and wavelength are unchanged.
- The **transmitted** wave keeps the same frequency but changes speed, so its wavelength changes (and it may change direction, which is refraction).

- The more different the two media are, the bigger the share that is reflected. Boundaries between similar materials send back only a faint echo.

The pulse-echo technique

Send out a **short pulse** of waves and time how long the echo takes to come back. If you know the wave speed in the medium, you know how far away the reflecting surface is. The pulse travels there *and back*, so the distance is half of speed × time:

Distance from a pulse-echo time

$$d = \frac{vt}{2}$$

d: distance to the reflector / m v: wave speed / m s^{-1}

t: time from sending the pulse to receiving its echo / s

This one idea runs **sonar** and ships' echo sounders (sound in water), **medical ultrasound** (each tissue boundary returns an echo, and the echo times give the depths), **radar** (radio waves off aircraft) and the way bats find insects in the dark.

What limits the information?

The spec asks you to explain two limits.

- **The duration of the pulse.** A pulse lasting a time τ is a train of waves of length $v\tau$. The echoes from two boundaries a distance Δd apart are separated in time by $2\Delta d/v$. If that is shorter than τ, the echoes overlap and merge into one, so you can't tell there were two boundaries. You can only separate reflectors further apart than about $v\tau/2$. **Shorter pulses give finer detail.**
- **The wavelength.** A wave can't pick out features much smaller than its wavelength: it diffracts round them instead of reflecting a clear echo. To see detail on a scale of 1 mm you need λ of about 1 mm or less, which is why medical ultrasound uses frequencies of several MHz. Higher frequency also helps the first limit, because a pulse has to contain at least one or two complete cycles, so short pulses need short periods.

NOT IN SPEC There is a catch: higher frequencies are absorbed more strongly by tissue, so they don't reach as deep, and scanners have to trade detail against depth. The pulses also have to be spaced out, because each echo must return before the next pulse is sent.

WORKED EXAMPLE

An ultrasound scanner sends pulses of frequency 3.0 MHz into soft tissue, where the speed of sound is 1540 m s^{-1}. Each pulse lasts 2.0 μs. An echo from a boundary returns 65 μs after the pulse is sent.
(a) Calculate the depth of the boundary.

(b) Calculate the wavelength and the number of complete waves in each pulse.
(c) Two boundaries are 1.0 mm apart. Explain whether this scanner can show them as separate boundaries.

Show solution

(a) The pulse goes there and back, so halve the total distance:

$$d = \frac{vt}{2} = \frac{1540 \times 65 \times 10^{-6}}{2} = 0.050 \text{ m} = 5.0 \text{ cm}$$

(b) $\lambda = \frac{v}{f} = \frac{1540}{3.0 \times 10^{6}} = 5.1 \times 10^{-4} \text{ m} = 0.51 \text{ mm}$. Number of cycles = pulse duration × frequency $= 2.0 \times 10^{-6} \times 3.0 \times 10^{6} = 6$.

(c) The pulse is $v\tau = 1540 \times 2.0 \times 10^{-6} = 3.1 \text{ mm}$ long. Echoes from boundaries 1.0 mm apart arrive $2 \times 1.0 \times 10^{-3}/1540 = 1.3 \ \mu\text{s}$ apart, which is less than the 2.0 μs pulse duration. The echoes overlap and merge, so the scanner **cannot** resolve them. It would need pulses shorter than 1.3 μs (for example fewer cycles, or a higher frequency).

EXAM TIP

In pulse-echo questions, **halve the time** (or the distance) before anything else. Examiners also like "explain why ultrasound uses such high frequencies": answer with wavelength (detail must be bigger than about λ) **and** pulse duration (short pulses need high frequencies).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Acoustic impedance: why the scan needs gel

How much of a sound wave reflects at a boundary depends on the **acoustic impedance** $Z = \rho v$ of each material (density × speed of sound). The fraction of the intensity reflected is

$$\frac{I_r}{I_0} = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2$$

For fat ($Z \approx 1.38 \times 10^{6} \text{ kg m}^{-2}\text{s}^{-1}$) next to muscle ($Z \approx 1.70 \times 10^{6}$) only about 1% reflects: enough for a faint echo, while 99% carries on to deeper layers. That is what makes imaging possible. But for air ($Z \approx 410$) next to skin ($Z \approx 1.6 \times 10^{6}$) the ratio is so extreme that about 99.9% reflects. Any air gap between the probe and the skin would bounce almost all the ultrasound straight back out. The gel has an impedance close to skin's, so it removes the air and lets the pulses in.

Quick checks

QUICK CHECK

In a sound wave, what is the distance between the centre of a compression and the centre of the next rarefaction?

- $\lambda/4$
- $\lambda/2$
- λ
- 2λ

Compression to compression is one wavelength, and the rarefaction sits halfway between them. On a transverse wave, crest to trough is $\lambda/2$ for the same reason.

QUICK CHECK

Sound passes from air into water, where it travels about four times faster. What happens to its frequency and wavelength?

- Frequency increases, wavelength stays the same
- Frequency stays the same, wavelength decreases
- Frequency stays the same, wavelength increases
- Both increase

The source fixes the frequency, so f can't change at the boundary. Since $v = f\lambda$, a four times bigger v means a four times longer λ.

QUICK CHECK

A transverse wave travels to the right along a rope. Point P has zero displacement, and the nearest crest is just to its left. Which way is P moving at this instant?

- [x] Upwards
- [ ] Downwards
- [ ] To the right
- [ ] It is momentarily at rest

The crest is moving right, towards P, so P is about to be at a crest: it must be rising. Particles in a transverse wave never move along the direction of travel, and they are only at rest at a crest or trough, not at zero displacement (where they move fastest).

QUICK CHECK

The intensity 2.0 m from a point source is 36 W m^{-2}. What is the intensity 6.0 m from the source?

- [ ] 12 W m^{-2}
- [ ] 6.0 W m^{-2}
- [x] 4.0 W m^{-2}
- [ ] 1.0 W m^{-2}

The distance has tripled, so the area has gone up by $3^2 = 9$: $36/9 = 4.0\ \mathrm{W\ m^{-2}}$. Dividing by 3 would treat intensity as $\propto 1/r$.

QUICK CHECK

A ship's echo sounder receives the echo from the sea bed 0.40 s after sending the pulse. Sound travels at 1500 m s^{-1} in sea water. How deep is the sea?

- 150 m
- 300 m
- 600 m
- 3750 m

$d = vt/2 = 1500 \times 0.40/2 = 300$ m. 600 m is the total distance down and back up.

QUICK CHECK

In Core Practical 4, the signal generator is set to 4.0 kHz. Through what distance must the microphone be moved for the two traces to go from in phase to in phase again? (Speed of sound = 340 m s^{-1}.)

- 4.3 cm
- 8.5 cm
- 17 cm
- 85 cm

The traces come back into phase each time the microphone moves one wavelength: $\lambda = v/f = 340/4000 = 0.085$ m. Half of this (4.3 cm) is where they are in antiphase.